

Abstract

Neuron level explanations assume that specific neurons consistently encode stable semantic features, making them useful for interpreting model behavior. However, the reliability of these activations across highly similar objective statements is underexamined, creating a gap in understanding whether neuron based explanations are truly trustworthy. If activations are inconsistent, then explanations built on them may be unstable or misleading. This study tests whether highly similar objective statements activate the same neurons. Using randomization tests, it evaluates whether observed activation overlap exceeds chance levels, challenging the assumption that neuron activations reliably represent semantic meaning. Using a pretrained transformer-based language model, we construct pairs of semantically similar but structurally varied objective statements. We record neuron activations using forward-pass hooks, and measure overlap using the Jaccard similarity of the top-k activated neurons and cosine similarity of the full activation vectors. Permutation-based randomization tests are used to generate a null distribution to determine whether observed overlap exceeds the overlap expected by random chance. Neuron activation overlap between similar objective statements does not significantly exceed overlap expected by random chance, with randomization test p-values above 0.05 for most comparisons. These results suggest that neuron-level explanations of semantic meaning are unreliable. Neural Networks have challenges when it comes to returning consistently explainable results, especially when using this technology for natural language processing. Analyzing models at the neuron level and trying to return explainable results becomes a challenge due to the uncertainty of results from neurons firing, even when using objective statements that should lead to consistency. This poses significant challenges when using this technology for highly-consequential decision making due to both the unreliability of the decisions being made, and the unreliability of explaining which neurons fired to make the decision. Users are more likely to face harm and less likely to be trusted if explanations are unclear and inconsistent from the black-box technology they use.

Sources

- Bau, D., Zhou, B., Khosla, A., Oliva, A., & Torralba, A. (2017). Network dissection: Quantifying interpretability of deep visual representations. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 6541–6549. <https://doi.org/10.1109/CVPR.2017.354>
- Cer, D., Diab, M., Agirre, E., Lopez-Gazpio, I., & Specia, L. (2017). SemEval-2017 Task 1: Semantic textual similarity multilingual and crosslingual focused evaluation. *Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017)*, 1–14. <https://doi.org/10.18653/v1/S17-2001>
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 4171–4186. <https://doi.org/10.18653/v1/N19-1423>
- Ganesh, Prakhar, et al. “On the Impact of Machine Learning Randomness on Group Fairness.” *ArXiv (Cornell University) FAccT '23: Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, 12 June 2023, <https://doi.org/10.1145/3593013.3594116>.
- Ghorbani, A., Abid, A., & Zou, J. (2019). Interpretation of neural networks is fragile. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01), 3681–3688. <https://doi.org/10.1609/aaai.v33i01.33013681>
- Hu, Qiang, et al. “Towards Understanding Model Quantization for Reliable Deep Neural Network Deployment.” *2023 IEEE/ACM 2nd International Conference on AI Engineering – Software Engineering for AI (CAIN)*, 1 May 2023, <https://doi.org/10.1109/cain58948.2023.00015>.
- Olah, C., Cammarata, N., Schubert, L., et al. (2020). Zoom in: An introduction to circuits. *Distill*. <https://distill.pub/2020/circuits/zoom-in/>
- Räuker, T., Ho, A., Casper, S., & Hadfield-Menell, D. (2023). Transparency illusions: The case of interpretability in machine learning. *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. <https://doi.org/10.1145/3593013.3594091>
- Tenney, I., Das, D., & Pavlick, E. (2019). BERT rediscovers the classical NLP pipeline. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 4593–4601. <https://doi.org/10.18653/v1/P19-1452>

Zini, Julia El, and Mariette Awad. "On the Explainability of Natural Language Processing Deep Models." *ACM Computing Surveys*, vol. 55, no. 5, 19 July 2022, <https://doi.org/10.1145/3529755>.